[26] On Large-Batch Training for Deep Learning: Generalization Gap and Sharp Minima

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RQ2 / RQ4 — batch size, sharpness, and the generalization gap.

Problem. Why can large-batch SGD yield worse test accuracy than small-batch SGD even when both achieve low training error? **Method.** The authors compare small- and large-batch training over several networks and tasks; they measure local sharpness of reached solutions and test strategies to reduce the gap. **Finding.** In their experiments, large-batch methods tend to reach sharper regions and generalize worse; small-batch methods find flatter regions. This is an empirical pattern, not a theorem that batch size uniquely determines generalization.

1. Batch gradients and their noise

With n samples and minibatch B_t of size b, SGD uses

$$g_t = \frac{1}{b}\sum_{i\in B_t}\nabla \ell_i(\theta_t), \qquad \theta_{t+1} = \theta_t - \eta g_t.$$

If samples are independent with per-example gradient covariance $\Sigma(\theta)$, the minibatch gradient covariance is approximately

$$\mathrm{Cov}(g_t \mid \theta_t) \approx \frac{\Sigma(\theta_t)}{b}.$$

Thus larger b lowers gradient noise. The exact finite-population variance also has a correction for sampling without replacement. Reduced noise is a plausible mechanism affecting which basin training explores, *not proof* that more noise always means a flatter or better solution.

2. Sharpness and the local Hessian

Around a local minimum θ^*, Taylor's theorem gives

$$L(\theta^* + u) - L(\theta^*) \approx \tfrac{1}{2}u^\top H u.$$

For $\|u\|_2 \le r$, the maximum quadratic increase is about

$$\max_{\|u\|\le r}\big(L(\theta^*+u) - L(\theta^*)\big) \approx \frac{1}{2}r^2\lambda_{\max}(H).$$

This is a helpful *Hessian interpretation* of sharpness. Keskar et al.'s operational sharpness measure instead uses the maximum observed objective increase over a local neighborhood (with coordinatewise scaling), not simply $\lambda_{\max}(H)$. Keep this distinction explicit when discussing their plots.

Teaching example. $L_1(x,y) = x^2 + 100y^2$ and $L_2(x,y) = x^2 + y^2$ both have zero training loss at the origin. But L_1 grows far faster for a small y displacement. Local geometry can differ despite matching training objectives; it alone does not determine test loss.

3. Empirical results and proposed explanations

The paper reports a small-batch/large-batch generalization gap across the tested tasks and relates it to the sharpness of the solutions found. It argues that minibatch noise may help avoid sharp minimizers. Its authors also explore practical measures to improve large-batch behavior, rather than claiming large-batch training is inherently doomed.

4. Limits, follow-up debate and survey use

Do not universalize. Batch size interacts with learning-rate schedule, training duration, normalization and model scale. Raw sharpness depends on how weights are parameterized; reparameterization counterexamples are an explicit counterpoint in our survey [27]. The results support a correlation in the studied settings, not a universal sharpness ⇒ generalization law.

Survey connection. RQ2: gradient-noise scale and basin selection; RQ4: empirical/theoretical mismatch. **Our Notes.** This paper popularized the idea that stochastic noise changes the minima we reach, not just the speed of

training. Its claim is important, but we need to read it alongside the reparameterization critique before drawing conclusions about real modern training.

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